

VINH, Viet Nam

WMO: 488450

Lat: 18.667N

Lon: 105.683E

Elev: 6

StdP: 101.25

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	12.4	7.5	6.4	14.2	9.1	7.2	14.4	5.3	20.0	4.5	19.5	0.9	320	0.263

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.4	37.2	27.0	36.2	26.8	35.1	26.7	28.4	33.3	28.0	32.6	27.6	32.1	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.2	30.7	26.9	22.6	30.3	26.6	22.1	30.0	92.2	33.5	90.1	32.7	88.4	32.3	34.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.0	4.8	4.1	DB	9.8	38.6	1.5	0.9	8.6	39.3	7.7	39.8	6.9	40.3	5.8	41.0	
			WB	8.7	30.0	1.6	1.7	7.5	31.2	6.6	32.2	5.7	33.2	4.6	34.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.0	18.2	19.1	21.4	25.5	28.9	30.9	30.5	29.6	27.9	25.7	22.9	19.4
	DBStd	5.24	3.28	3.78	3.25	3.19	2.76	2.10	2.00	1.74	1.92	1.97	2.70	2.94
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	112	44	34	9	1	0	0	0	0	0	0	2	23
	CDD10.0	5490	254	256	354	466	587	628	636	607	538	488	386	291
	CDD18.3	2561	40	56	105	217	328	378	378	348	288	229	138	55
	CDH23.3	26722	52	154	391	1759	3935	5251	5113	4384	3101	1784	672	127
Wind	CDH26.7	11761	4	27	87	563	1837	2910	2690	2017	1120	398	102	7
	WSAvg	1.3	1.0	1.0	1.1	1.2	1.4	1.8	1.7	1.4	1.2	1.2	1.1	1.1
	PrecAvg	1998	48	38	55	63	164	104	135	245	451	453	163	80
	PrecMax	2638	115	69	143	169	423	316	379	716	941	1199	462	249
	PrecMin	1238	0	16	14	14	37	4	24	54	134	47	34	28
	PrecStd	364	27	16	28	35	88	83	77	164	212	310	111	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	30.3	33.9	37.3	38.4	38.2	38.2	36.9	35.4	31.7	30.4	27.6
		MCWB	22.5	23.2	25.3	27.0	27.6	27.4	26.9	27.0	27.0	26.1	25.9	23.9
	2%	DB	24.6	26.5	28.7	34.0	36.8	37.0	36.6	35.3	33.8	30.6	29.0	26.0
		MCWB	21.7	23.2	24.9	27.0	27.1	27.0	26.7	26.8	27.0	25.5	24.9	22.9
	5%	DB	23.5	25.0	26.7	31.3	35.1	35.9	35.4	34.1	32.4	29.7	27.8	24.7
		MCWB	21.2	22.6	24.0	26.6	26.8	26.8	26.6	26.6	26.7	25.2	24.2	22.3
	10%	DB	22.4	23.9	25.2	29.6	33.4	34.8	34.2	33.0	31.3	28.9	26.6	23.5
		MCWB	20.6	22.1	23.2	25.9	26.9	26.6	26.5	26.6	26.3	24.9	23.6	21.5
	0.4%	WB	23.2	24.5	26.4	28.3	29.1	29.1	28.7	28.5	28.4	27.2	26.4	24.7
		MCDB	25.3	27.4	30.2	33.9	34.6	34.4	33.5	33.1	32.5	30.1	29.2	26.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	22.3	23.6	25.0	27.4	28.2	28.2	28.0	27.9	27.7	26.5	25.6	23.7
		MCDB	23.9	25.5	27.9	32.2	33.1	33.1	32.9	32.1	31.4	29.0	27.9	25.3
	5%	WB	21.6	23.0	24.3	26.7	27.7	27.8	27.5	27.5	27.3	26.0	25.1	22.7
		MCDB	23.1	24.7	26.4	30.7	32.4	32.5	32.3	31.6	30.8	28.4	27.0	24.2
	10%	WB	20.9	22.3	23.5	26.1	27.3	27.4	27.1	27.1	26.8	25.6	24.4	21.9
		MCDB	22.2	23.8	25.2	29.3	31.7	32.1	31.8	31.1	30.1	27.8	26.2	23.4
	5% DB	MDBR	4.2	4.2	4.5	5.4	6.4	6.4	6.2	5.9	5.4	4.8	4.8	4.5
		MCDBR	5.6	6.0	6.6	8.4	8.8	7.8	7.8	7.6	7.2	6.1	6.2	5.8
		MCWBR	2.8	2.8	3.0	3.4	2.6	2.3	2.3	2.4	2.5	2.0	2.5	2.9
		MCWBR	5.0	5.5	6.4	7.4	7.5	7.0	6.9	6.6	6.4	5.1	5.5	5.1
Clear-Sky Solar Irradiance	taub		0.582	0.619	0.731	0.682	0.532	0.472	0.450	0.491	0.562	0.659	0.597	0.628
		taud	1.813	1.772	1.591	1.721	2.127	2.310	2.366	2.237	2.008	1.738	1.832	1.726
	Ebn at Noon		699	701	642	682	784	824	844	816	756	662	681	642
		Edh at Noon	202	221	273	241	159	131	124	142	177	225	197	215
All-Sky Solar Radiation	RadAvg		2.54	3.12	3.41	4.46	5.17	5.18	4.95	4.67	4.16	3.45	3.05	2.37
	RadStd		0.47	0.47	0.34	0.40	0.46	0.38	0.59	0.35	0.47	0.47	0.41	0.38

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.43	N/A	N/A	+0.86	+0.31	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.33	N/A	N/A	+0.44	+0.20	+0.16	N/A	N/A	N/A	N/A

Nomenclature: See separate page